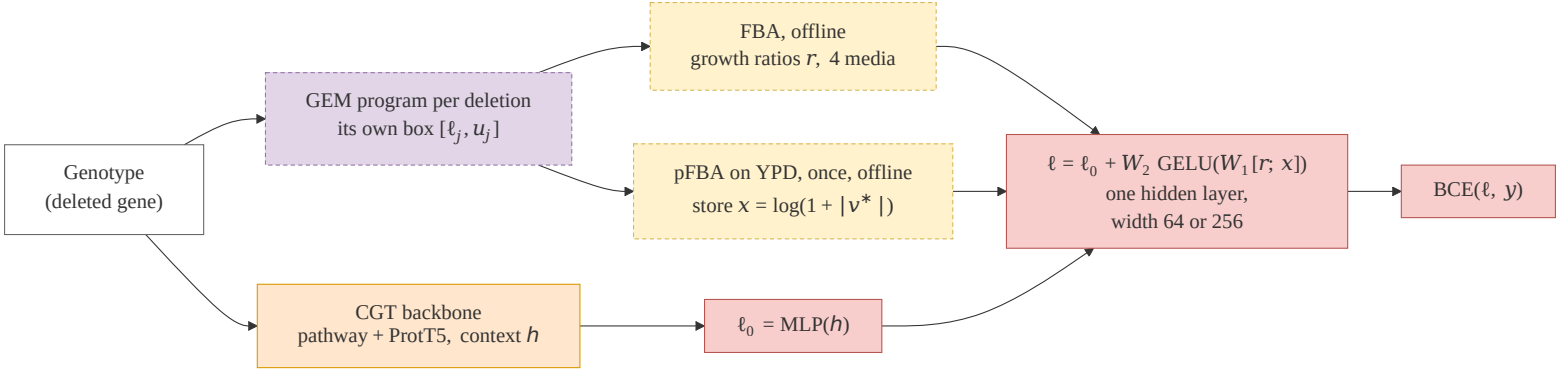

Genotype
(deleted gene)
GEM program per deletion
its own box $[\ell_j, u_j]$
FBA, offline
growth ratios r, 4 media
pFBA on YPD, once, offline
store $x = \log(1 + |v^*|)$
CGT backbone
pathway + ProtT5, context h
$\ell_0 = \mathrm{MLP}(h)$
$\ell = \ell_0 + W_2\ \mathrm{GELU}(W_1[r;\ x])$
one hidden layer,
width 64 or 256
BCE(ℓ, y)